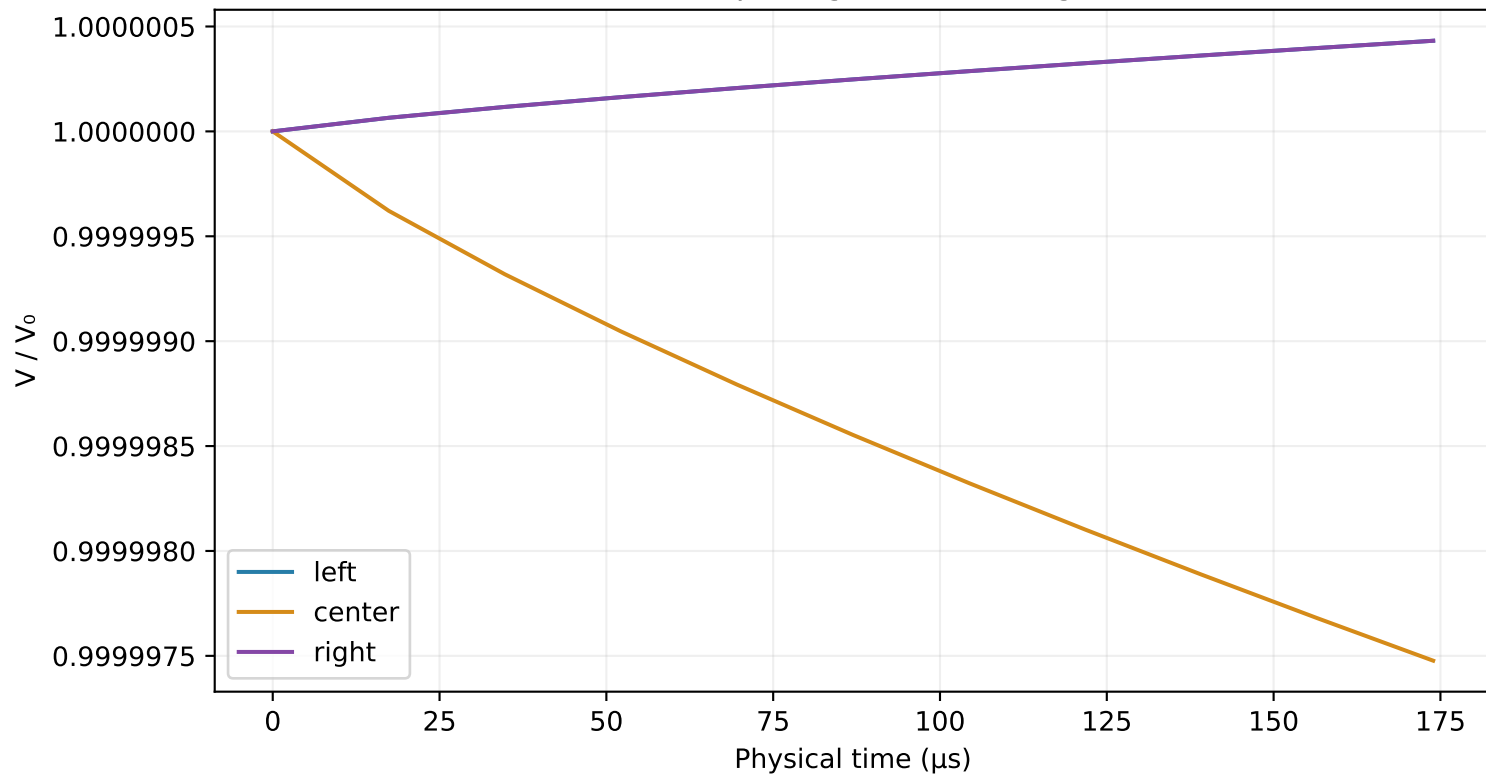
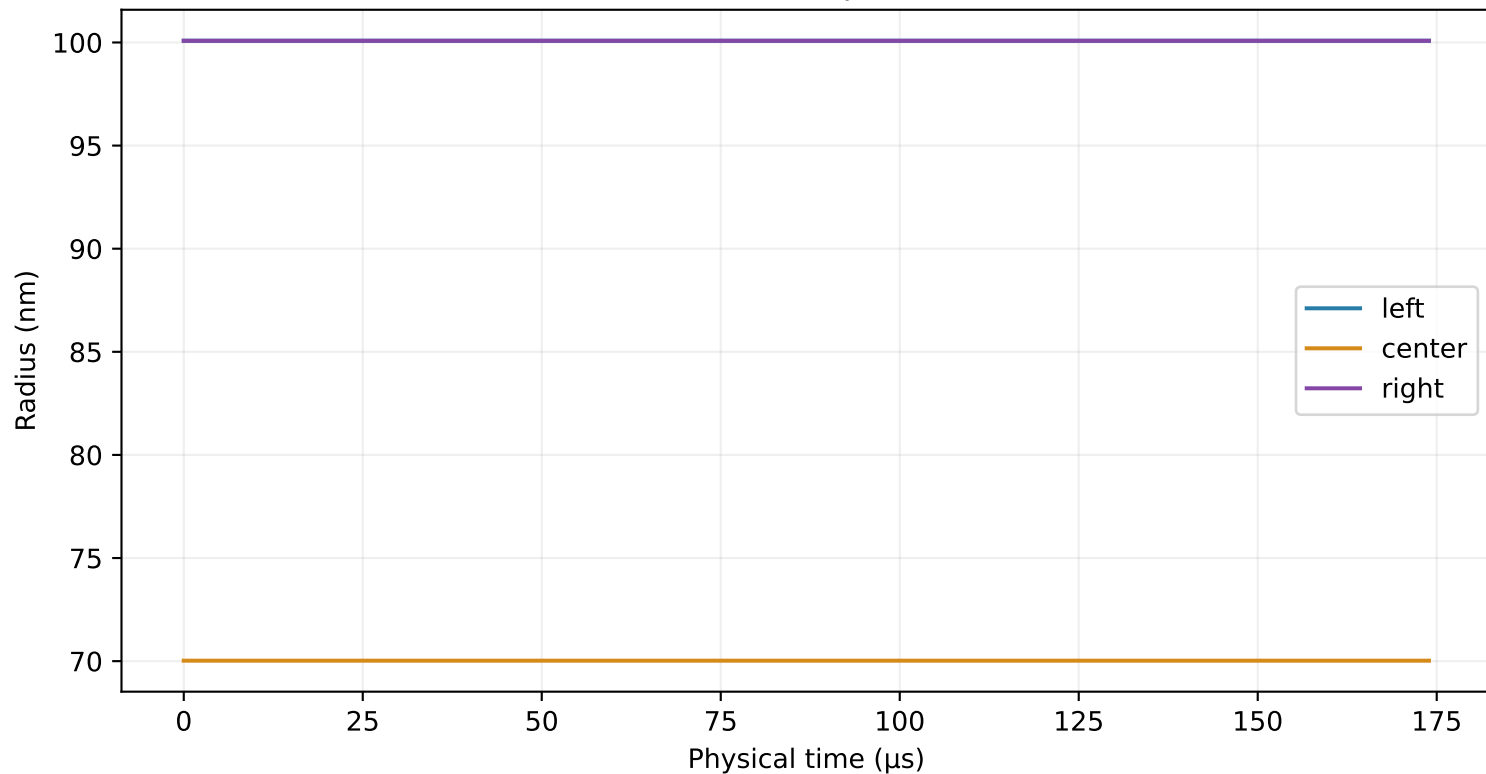


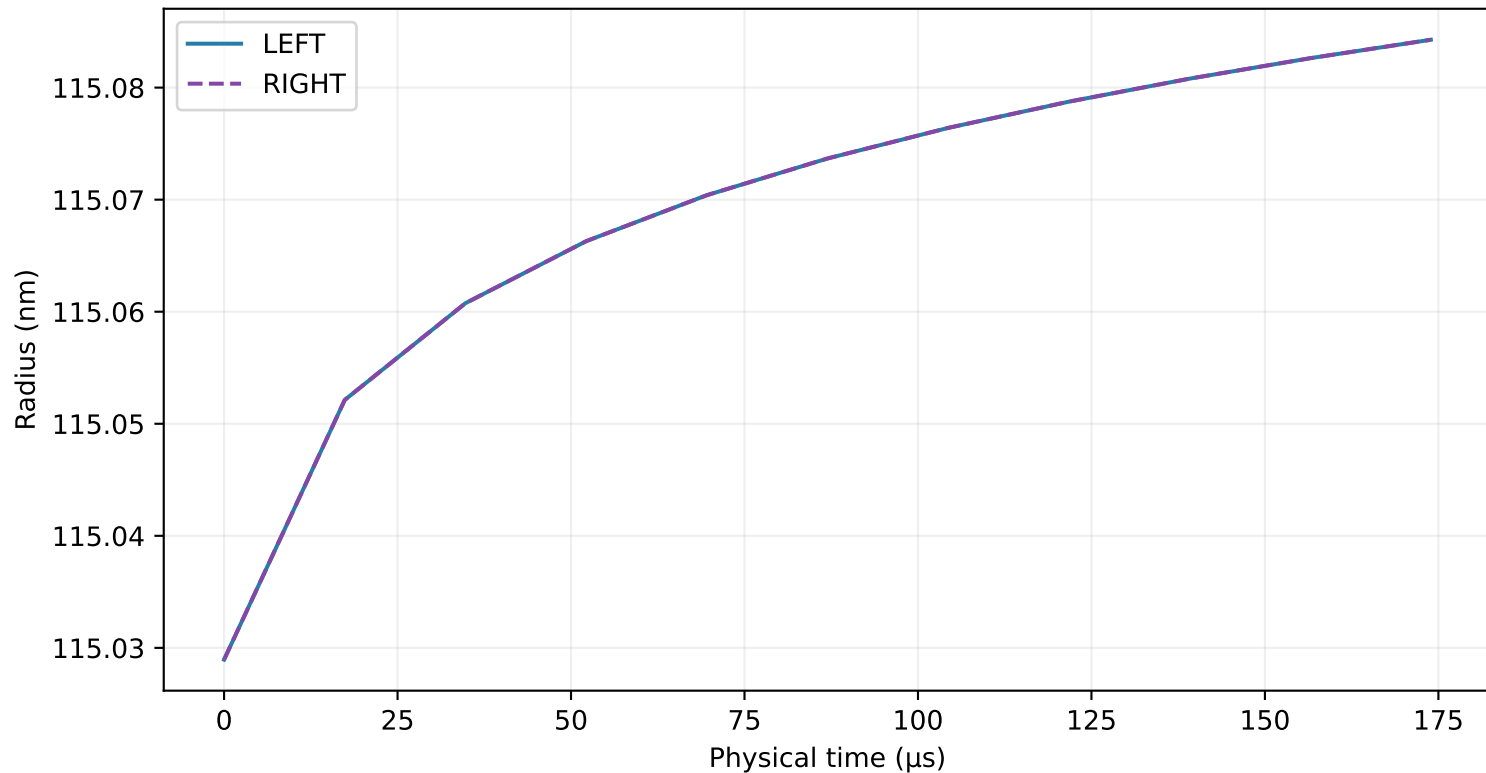
Native PF surface ripening: normalized grain volumes



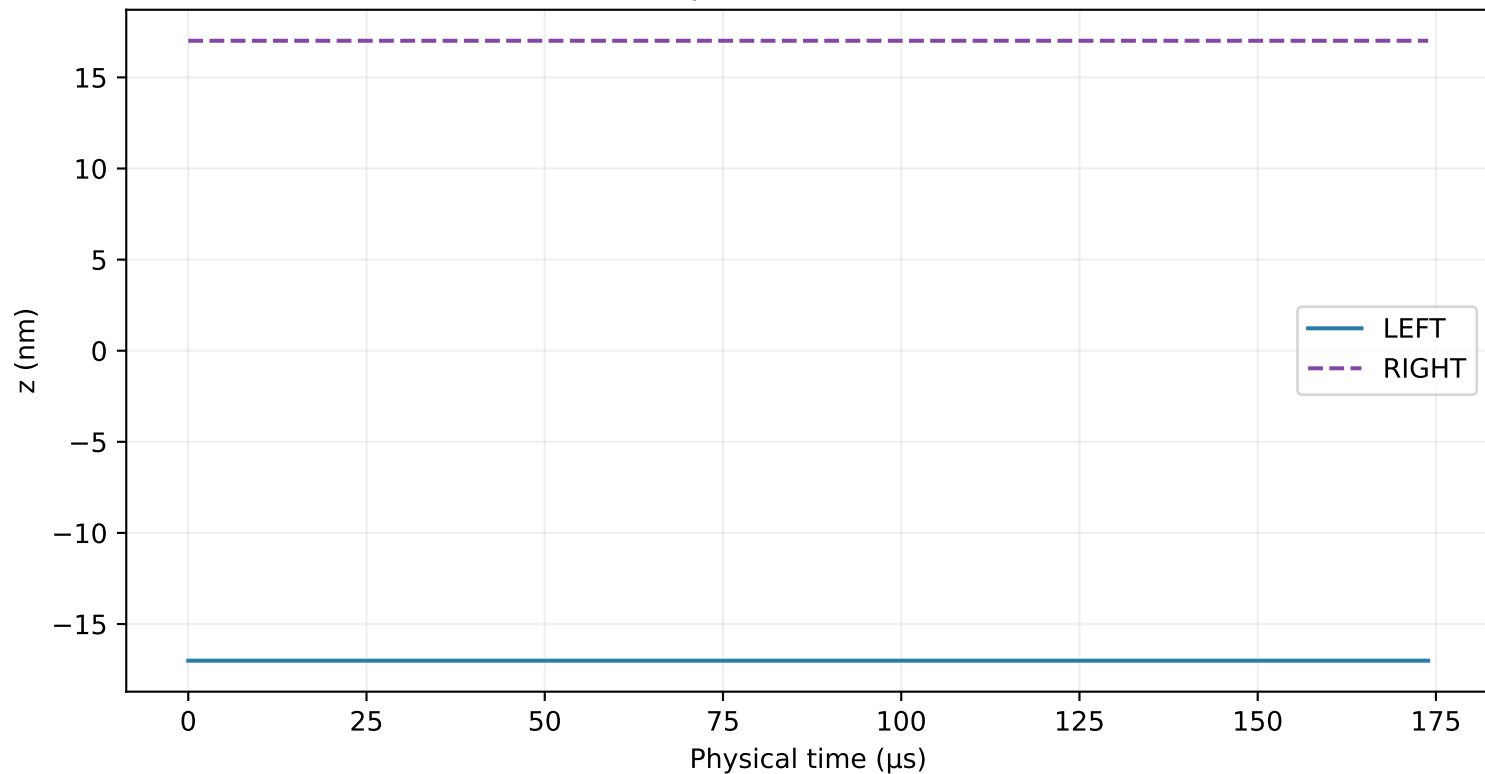
Current volume-equivalent radii



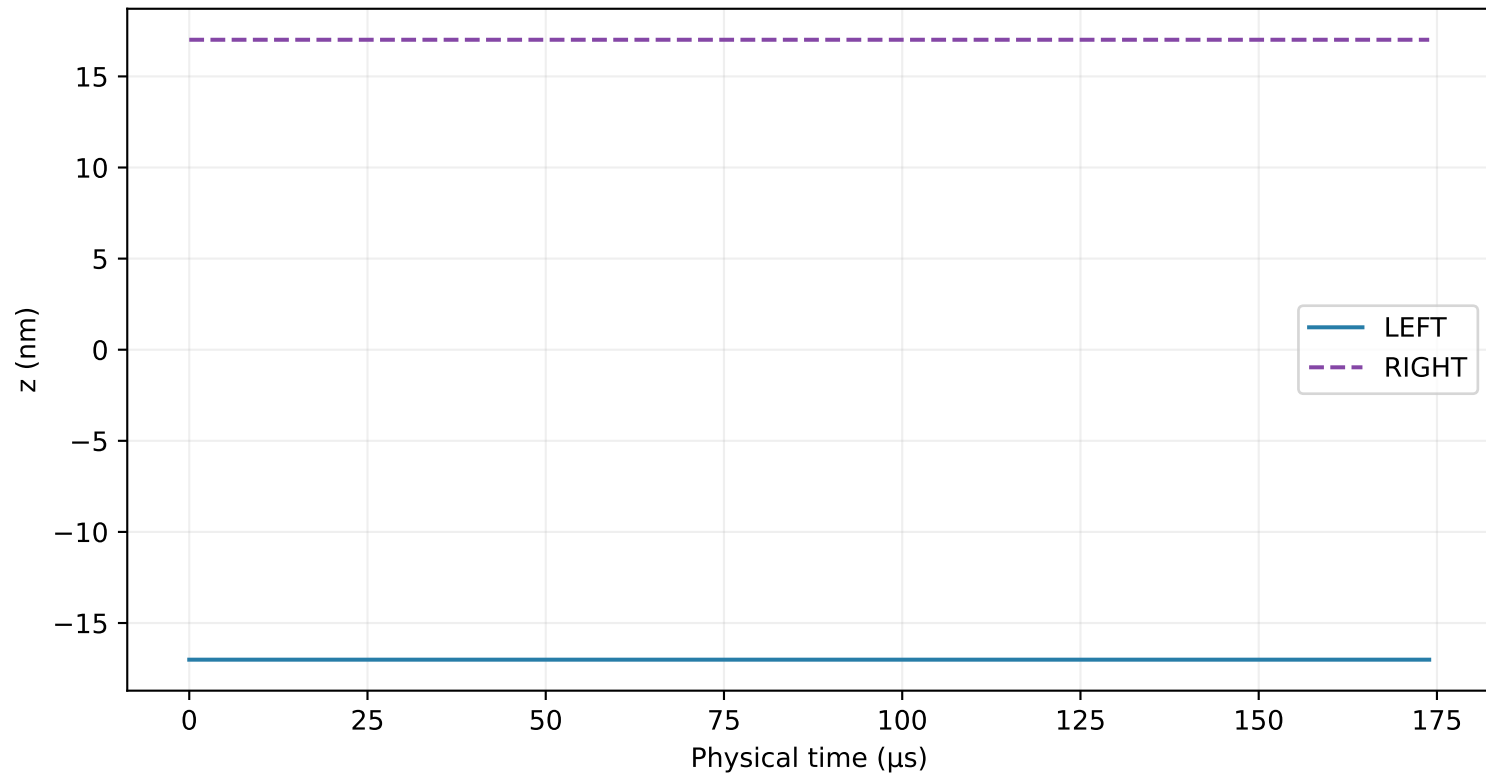
Neck Radii



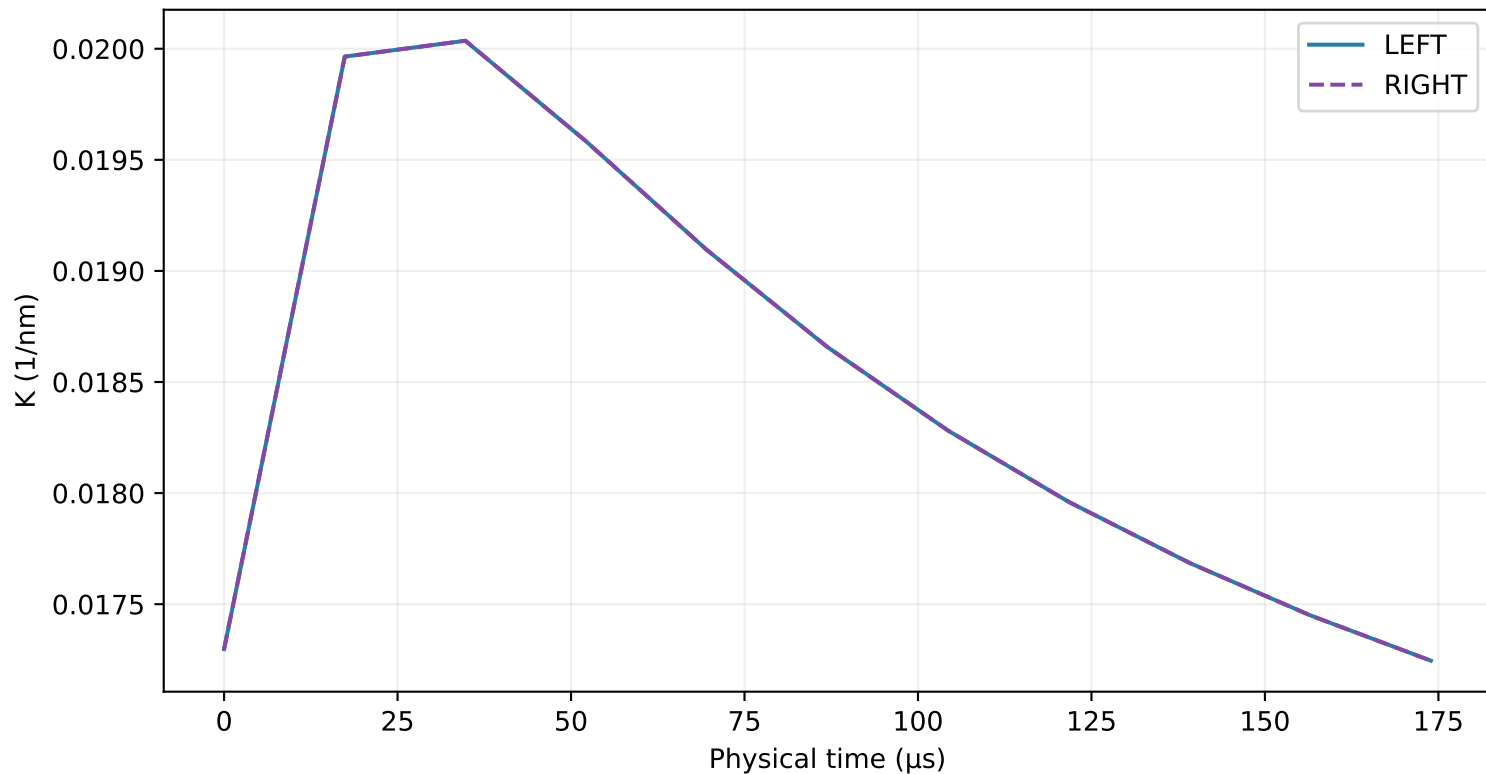
Tj Axial Positions



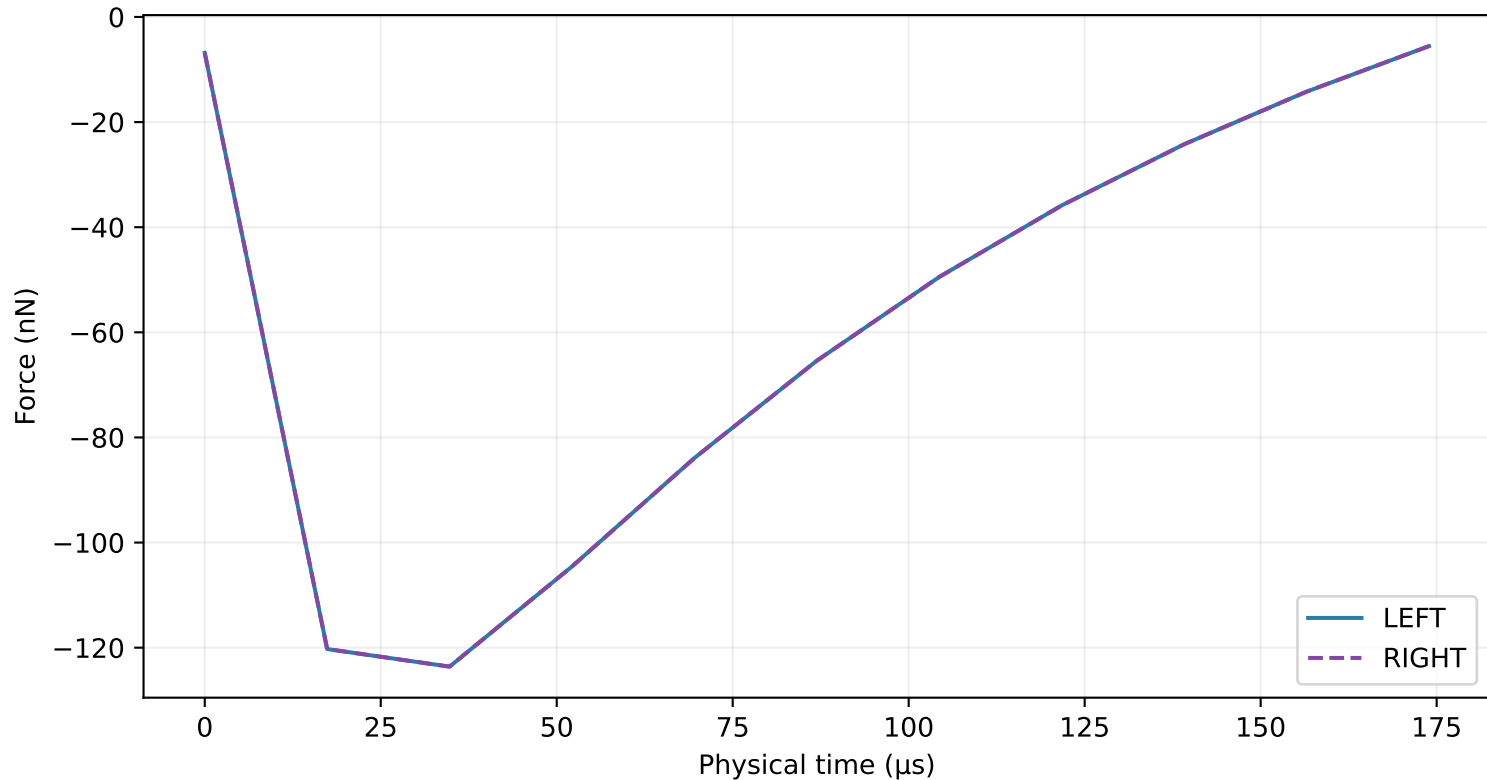
Gb Positions



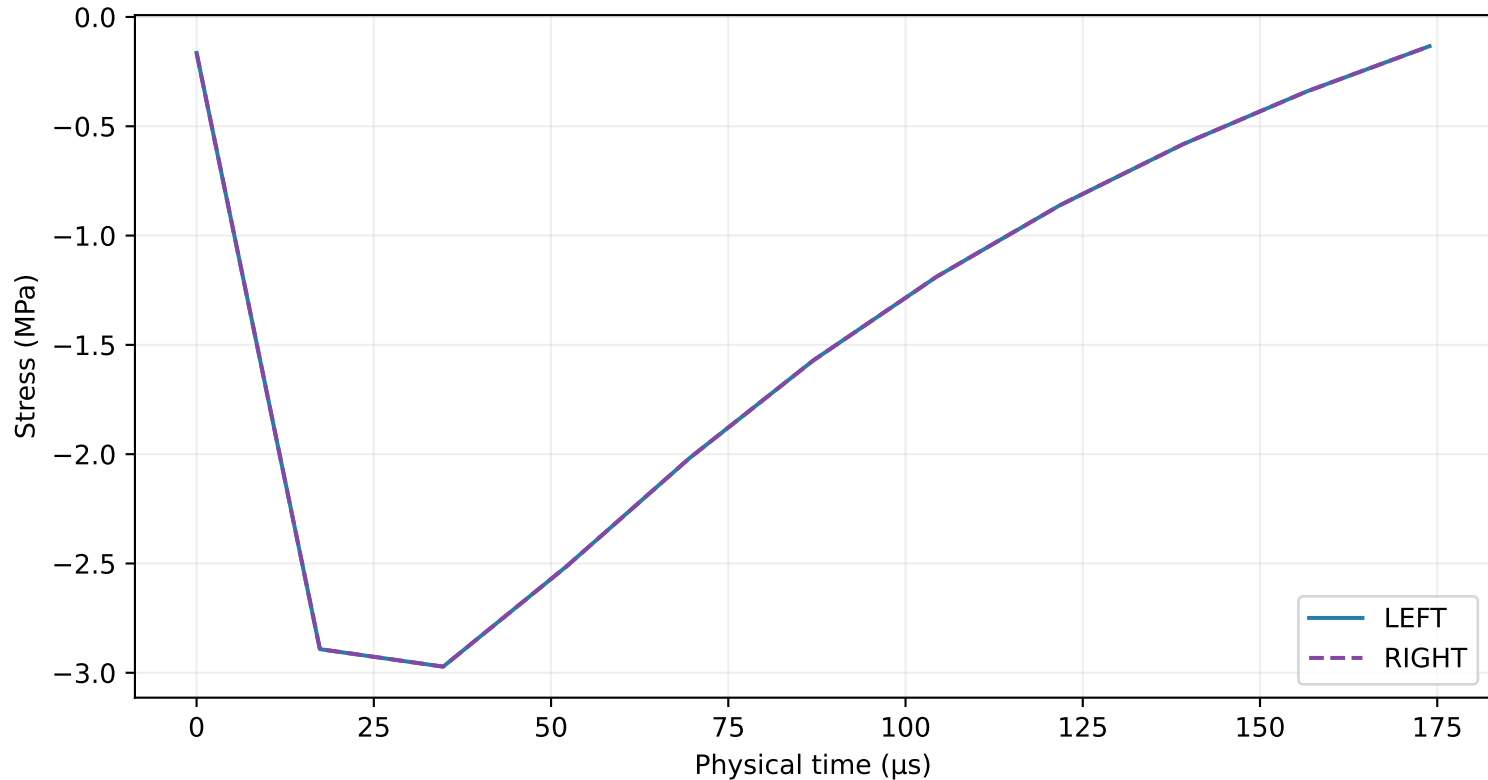
Neck Curvature



Cannon Carter Force

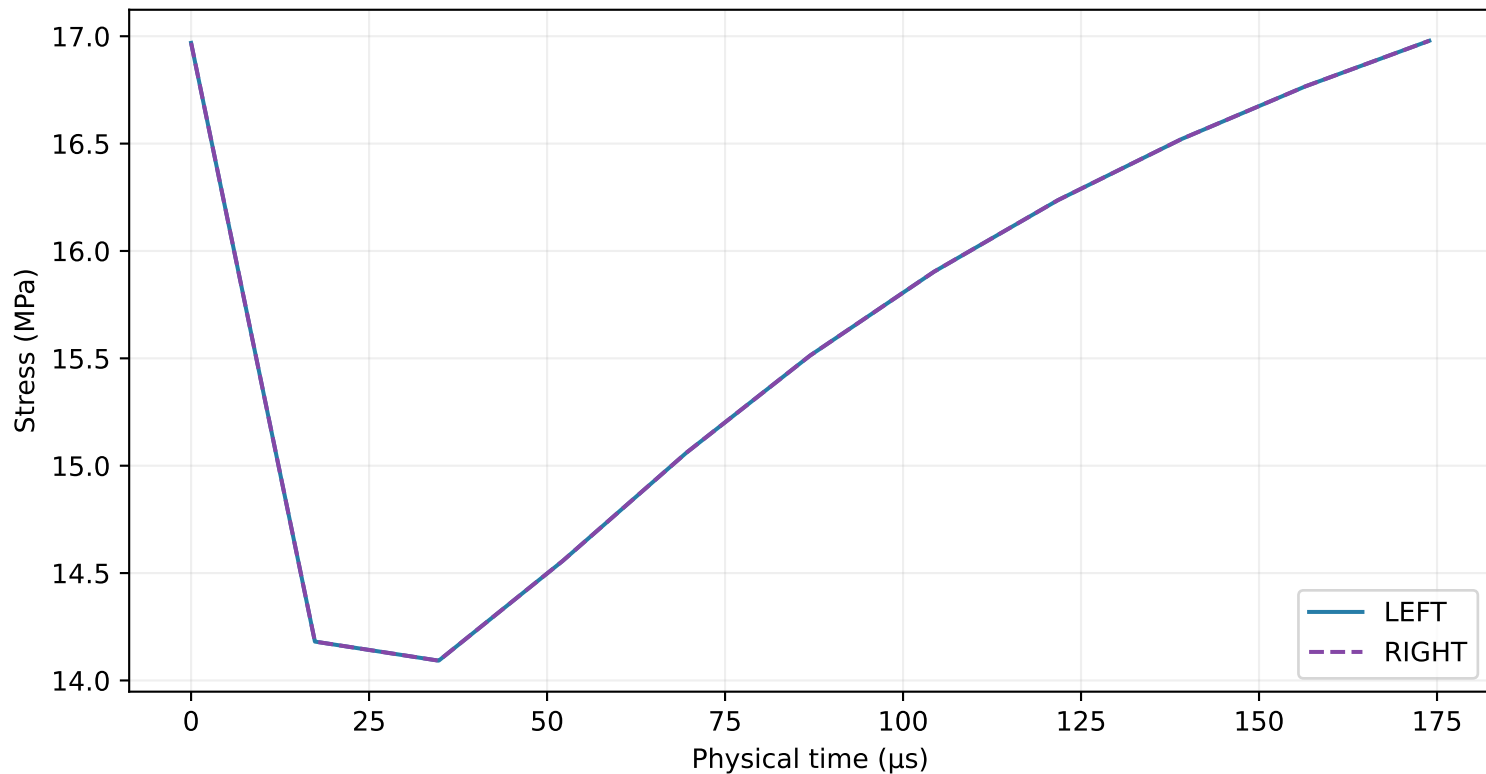


Cannon Carter Stress

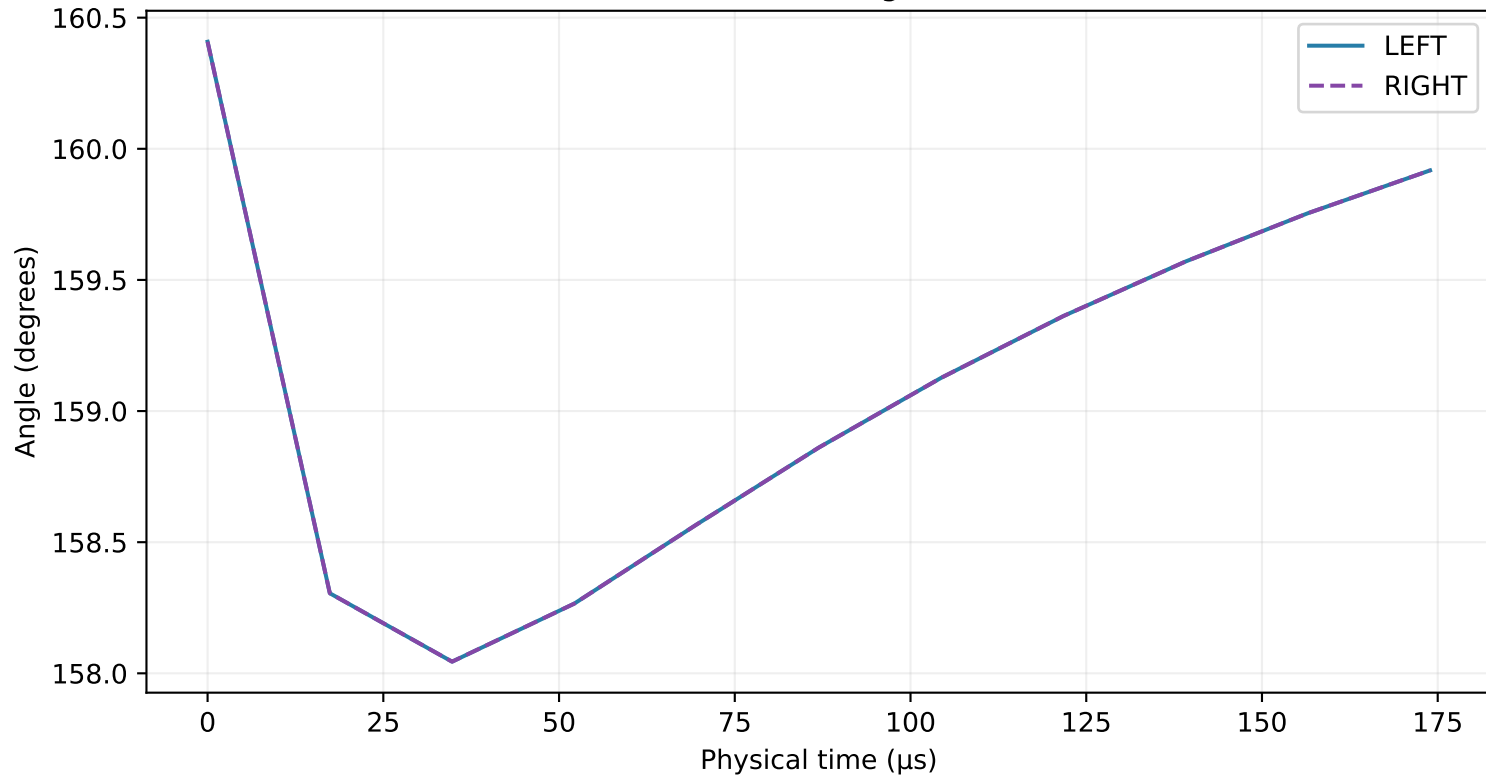




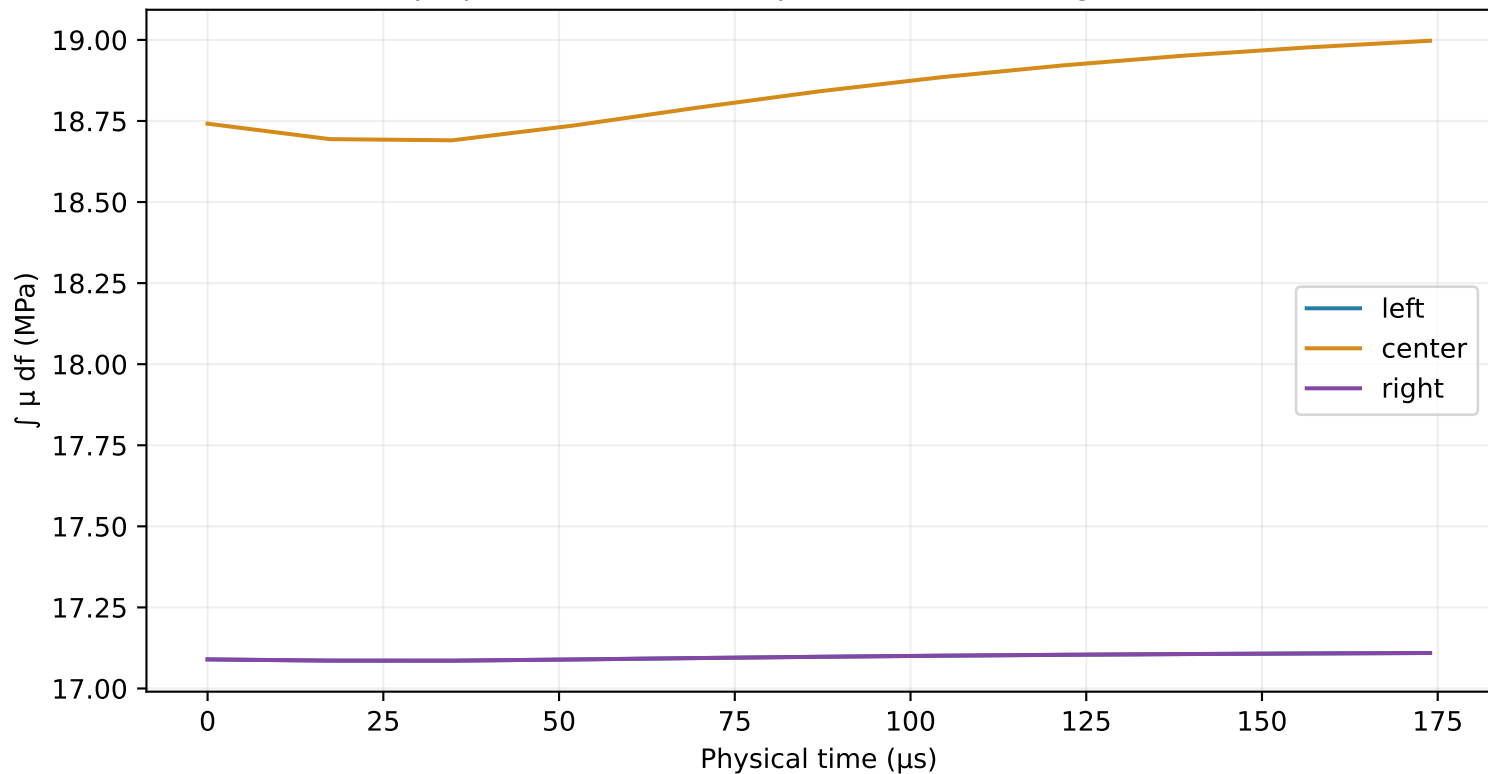
Pf Geometric Stress



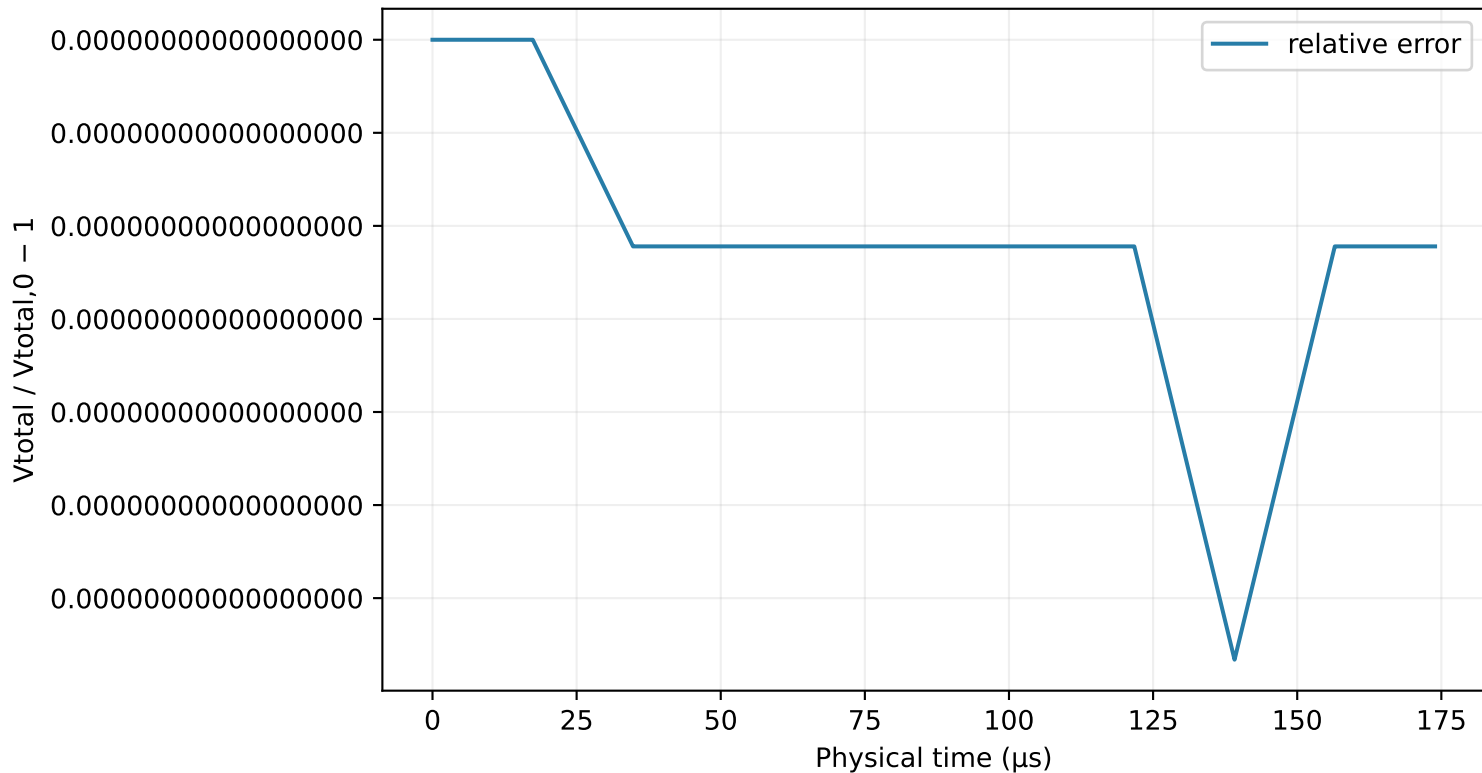
Dihedral Angles



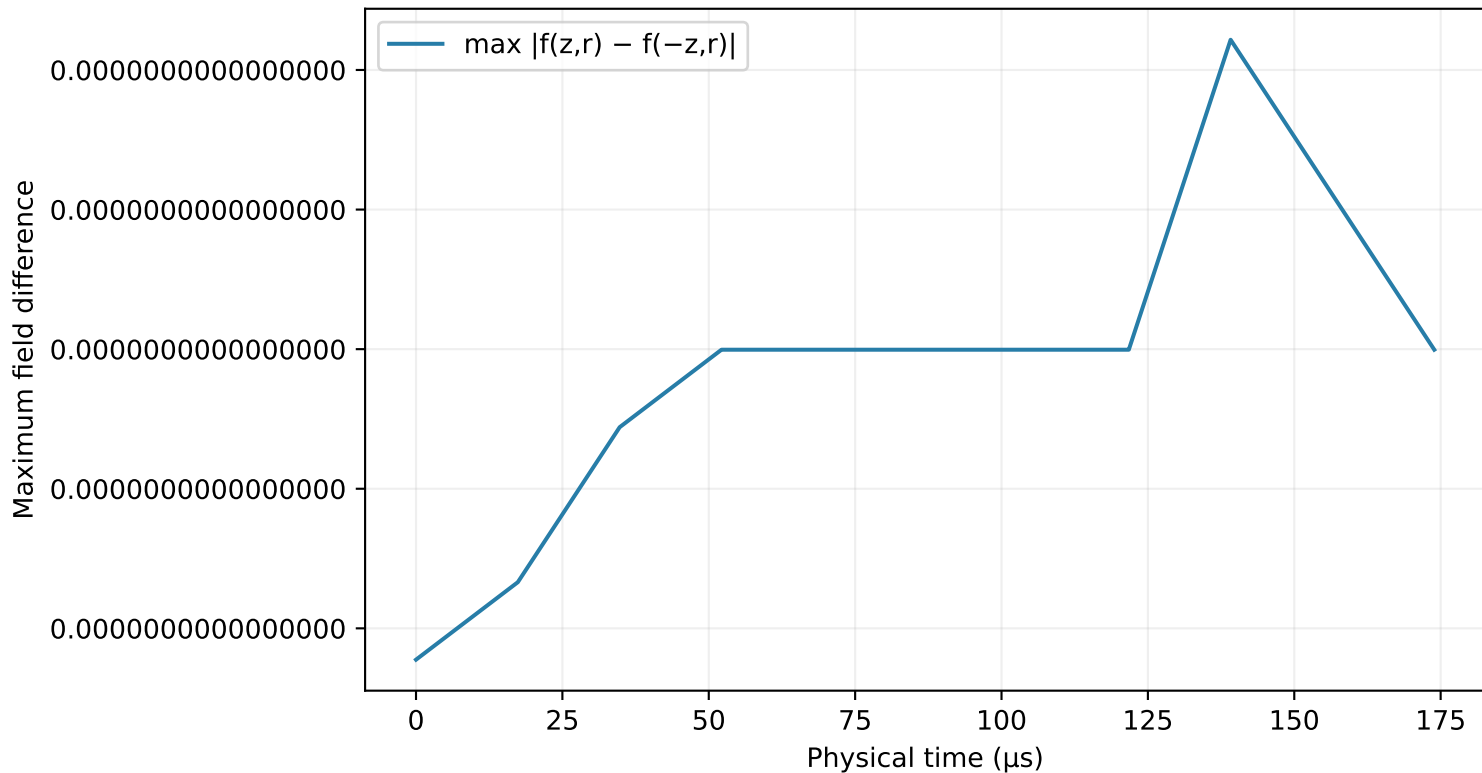
Normal-projected PF chemical potential, excluding 2W around GBs



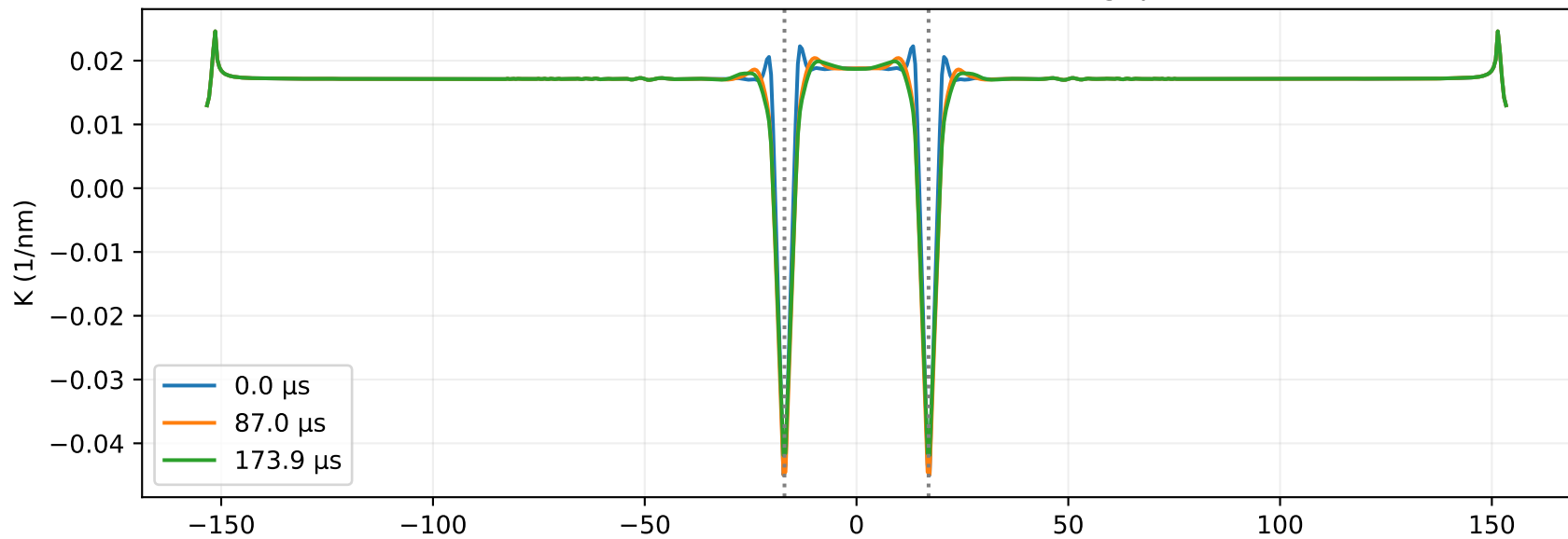
Global material conservation



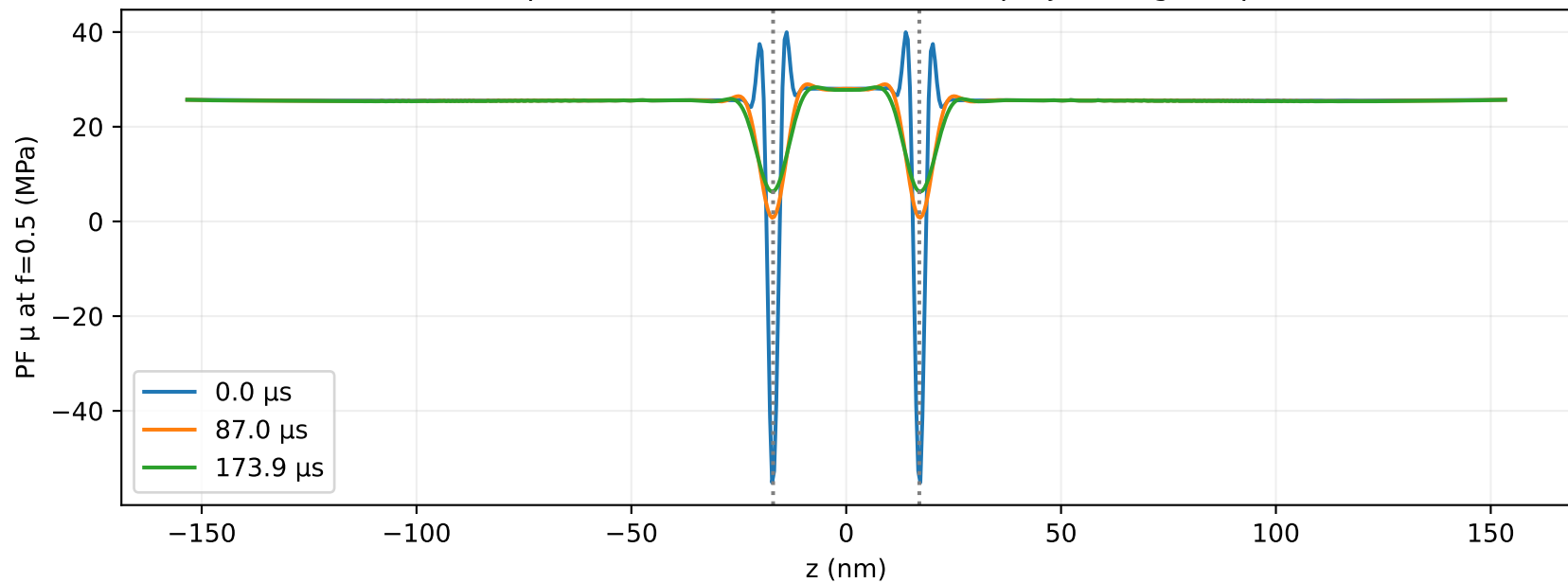
Mirror symmetry before any stochastic event



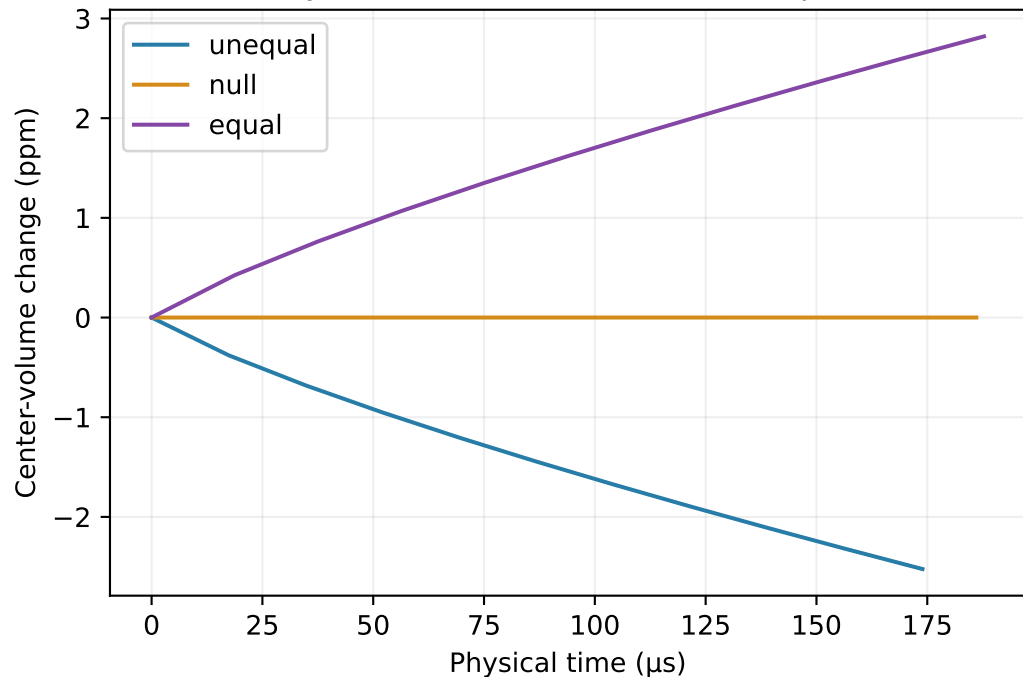
Full free-surface curvature (fixed W-scale smoothing; poles excluded)



Local PF field potential; distinct from normal-projected grain potential



Primary null vs size/coordination comparisons



No fitted transfer or sign correction

